

Project Title: PhilBot - Maneha Tamsaal

1. Literature Review & Data Understanding

1.1 Related Work (2–3 sources)

Source 1: UNESCO Recommendation on the Ethics of Artificial Intelligence

<https://unesdoc.unesco.org/ark:/48223/pf0000380455/>

- **What it does:** Offers a consistent system of global standards to solve ethical issues in AI, including gender discrimination and judicial accountability.
- **Relevance:** It gives a basis to the real-world issues the chatbot will solve especially in the area of technology and social ethics.

Source 2: Moral Dilemma Datasets (e.g., Moral Machine)

<https://www.moralmachine.net/>

<https://www.ethicsnet.org/datasets/>

- **What it does:** These are edited sets of situations, where there are competing moral values, like truth vs. loyalty, usually accompanied by human evaluations or explanations.
- **Relevance:** These datasets provide the uncoded "dilemma" content which our chatbot is able to rely on to initiate user interaction and philosopher "reactions" with.

1.2 Dataset Description

- **Dataset Source:**

Public domain philosophical texts (Project Gutenberg) and the EthicsNet or Moral Dilemma Datasets.

- **Type of Data (e.g., text, tabular, images):**

Primarily textual data consisting of historical philosophical treatises and structured moral vignettes.

- **Key Features:**

- Chatbot enabled design which makes learning philosophers easier, more humanized and less dense.
- Related Resources readily available and quotes.

- **Any assumptions / limitations:** Obviously a chatbot cannot bring the complete nuance of an elaborate text in a short conversation so it might limit the complete understanding of a philosophical framework.

2. Project Design Proposal

2.1 Problem Formulation

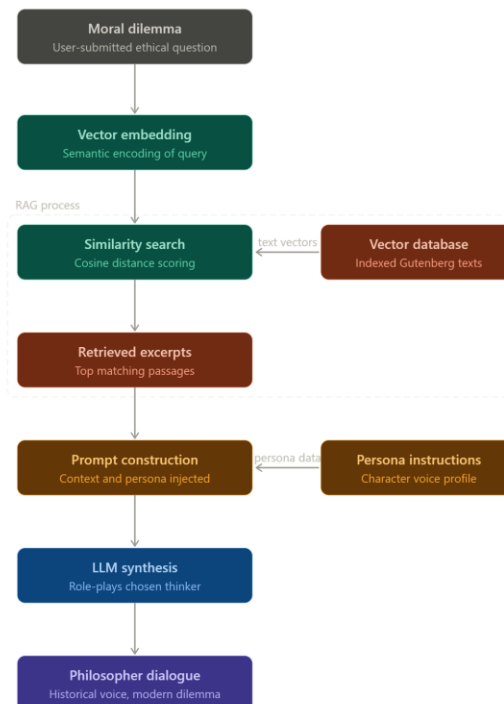
Traditional philosophy is often dense and daunting for beginners. We aim to utilize Natural Language Processing (NLP) to address this gap, to convert the static philosophical theories into an interactive conversational agent that can simulate certain personas to address modern ethical questions.

2.2 Proposed Approach

We will use Large Language Model (LLM) Fine-tuning or Prompt Engineering (Role-playing) combined with Retrieval-Augmented Generation (RAG). Allowing the bot to fetch specific quotes or concepts from a database of philosophical texts, thereby ensuring the "philosopher" stays true to their historical views while answering modern dilemmas.

2.3 System Workflow / Pipeline

The PhilBot pipeline begins when a user submits a moral dilemma, which is immediately converted into a vector embedding to capture its semantic meaning. This query is then sent to a Vector Database—containing indexed philosophical texts from Project Gutenberg—where a Retrieval-Augmented Generation (RAG) process identifies the most relevant excerpts based on mathematical similarity. These specific quotes are injected into a structured prompt along with "persona" instructions, guiding a Large Language Model (LLM) to synthesize a response. The final output is a conversational dialogue where the AI "acts" as the philosopher, using the retrieved historical evidence to address the modern ethical problem.



2.4 Algorithm / Logic Overview

The core logic of the system relies on Semantic Search and Contextual Prompt Engineering to ensure accuracy and character depth. By using embedding models like BERT or OpenAI's text-embedding-3, the system calculates the Cosine Similarity between the user's query and the philosophical database, ensuring that even if a user uses modern slang, the bot can map it to classical concepts like the "Categorical Imperative" or "Utilitarianism." The "Logic Gate" of the RAG framework acts as a constraint on the LLM, forcing it to prioritize the retrieved text over its general training data, which effectively prevents "hallucinations" and ensures the "philosopher" remains historically grounded while role-playing.

3. Design Reflection

3.1 Discussion (if any)

- I didn't have the chance to discuss with TA/LE but I did go over the project with some philosophy majors who helped me understand the importance of clarity and nuance I must bring to the project.

3.2 Feedback & Improvements

- Received feedback to include verified citations so the app remains an educational tool rather than just a chatbot.

4. Challenges Faced

- **Nuance:** Capturing the authentic "voice" of a philosopher without making it a caricature.
- **Bias:** Making sure that the datasets on which the model is being trained do not have contemporary biases that rewrite the desired philosophical view.

5. Next Steps

The next phase involves preprocessing the text data (cleaning and normalizing) and setting up a basic vector database to store philosophical texts for the RAG pipeline.